tinytable

Easy, beautiful, and customizable tables in R

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1 Tiny Tables

`tinytable` is a small but powerful `R` package to draw HTML, LaTeX, Word, PDF, Markdown, and Typst tables. The interface is minimalist, but it gives users direct and convenient access to powerful frameworks to create endlessly customizable tables.

Install the latest version from R-Universe or CRAN:

```
install.packages("tinytable",
    repos = c("https://vincentarelbundock.r-universe.dev", "https://cran.r-project.org"))
```

This tutorial introduces the main functions of the package. It is also available as a single PDF document.

Load the library and set some global options:

```
library(tinytable)
options(tinytable_tt_digits = 3)
options(tinytable_theme_placement_latex_float = "H")
```

Draw a first table:

```
x <- mtcars[1:4, 1:5]
tt(x)
```

mpg	cyl	disp	hp	drat
21	6	160	110	3.9
21	6	160	110	3.9
22.8	4	108	93	3.85
21.4	6	258	110	3.08

1.1 Width

The `width` arguments indicating what proportion of the line width the table should cover. This argument accepts a number between 0 and 1 to control the whole table width, or a vector of numeric values between 0 and 1, representing each column.

```
tt(x, width = 0.5)
```

mpg	cyl	disp	hp	drat
21	6	160	110	3.9
21	6	160	110	3.9
22.8	4	108	93	3.85
21.4	6	258	110	3.08

```
tt(x, width = 1)
```

mpg	cyl	disp	hp	drat
21	6	160	110	3.9
21	6	160	110	3.9
22.8	4	108	93	3.85
21.4	6	258	110	3.08

We can control individual columns by supplying a vector. In that case, the sum of `width` elements determines the full table width. For example, this table takes 70% of available width, with the first column 3 times as large as the other ones.

```
tt(x, width = c(.3, .1, .1, .1, .1))
```

mpg	cyl	disp	hp	drat
21	6	160	110	3.9
21	6	160	110	3.9
22.8	4	108	93	3.85
21.4	6	258	110	3.08

When the sum of the `width` vector exceeds 1, it is automatically normalized to full-width. This is convenient when we only want to specify column width in relative terms:

```
tt(x, width = c(3, 2, 1, 1, 1))
```

mpg	cyl	disp	hp	drat
21	6	160	110	3.9
21	6	160	110	3.9
22.8	4	108	93	3.85
21.4	6	258	110	3.08

When specifying a table `width`, the text is automatically wrapped to appropriate size:

```
lorem <- data.frame(
  Lorem = "Sed ut perspiciatis unde omnis iste natus error sit voluptatem accusantium dolorem
  Ipsum = " Nemo enim ipsam voluptatem quia voluptas sit aspernatur aut odit aut fugit, sed q
)

tt(lorem, width = 3/4)
```

Lorem	Ipsum
Sed ut perspiciatis unde omnis iste natus error sit voluptatem accusantium doloremque laudantium, totam rem aperiam, eaque ipsa quae ab illo inventore veritatis et quasi architecto beatae vitae dicta sunt explicabo.	Nemo enim ipsam voluptatem quia voluptas sit aspernatur aut odit aut fugit, sed quia consequuntur magni dolores eos.

1.2 Footnotes

The `notes` argument accepts single strings or named lists of strings:

```
n <- "Fusce id ipsum consequat ante pellentesque iaculis eu a ipsum. Mauris id ex in nulla co

tt(lorem, notes = n, width = 1)
```

Table 1: A full-width table with wrapped text in cells and a footnote.

Lorem	Ipsum
Sed ut perspiciatis unde omnis iste natus error sit voluptatem accusantium doloremque laudantium, totam rem aperiam, eaque ipsa quae ab illo inventore veritatis et quasi architecto beatae vitae dicta sunt explicabo.	Nemo enim ipsam voluptatem quia voluptas sit aspernatur aut odit aut fugit, sed quia consequuntur magni dolores eos.

Fusce id ipsum consequat ante pellentesque iaculis eu a ipsum. Mauris id ex in nulla consectetur aliquam. In nec tempus diam. Aliquam arcu nibh, dapibus id ex vestibulum, feugiat consequat erat. Morbi feugiat dapibus malesuada. Quisque vel ullamcorper felis. Aenean a sem at nisi tempor pretium sit amet quis lacus.

When `notes` is a named list, the names are used as identifiers and displayed as superscripts:

```
tt(x, notes = list(a = "Blah.", b = "Blah blah."))
```

mpg	cyl	disp	hp	drat
21	6	160	110	3.9
21	6	160	110	3.9
22.8	4	108	93	3.85
21.4	6	258	110	3.08

[a] Blah.
[b] Blah blah.

We can also add markers in individual cells by providing coordinates:

```
tt(x, notes = list(
    a = list(i = 0:1, j = 1, text = "Blah."),
    b = "Blah blah."
  )
)
```

Table 2: Blah blah blah

mpg	cyl	disp	hp
21	6	160	110
21	6	160	110
22.8	4	108	93
21.4	6	258	110

mpg[a]	cyl	disp	hp	drat
21 [a]	6	160	110	3.9
21	6	160	110	3.9
22.8	4	108	93	3.85
21.4	6	258	110	3.08

[a] Blah.
[b] Blah blah.

1.3 Captions and cross-references

In Quarto, one should always specify captions cross-references using chunk options, and should *not* use the `caption` argument. This is because Quarto automatically post-processes tables, and may introduce conflict with the captions inserted by `tinytable`. For example:

```
@tbl-blah shows that...

```{r}
#| label: tbl-blah
#| tbl-cap: "Blah blah blah"
library(tinytable)
tt(mtcars[1:4, 1:4])
```
```

And here is the rendered version of the code chunk above:

Table 2 shows that…

```
library(tinytable)
tt(mtcars[1:4, 1:4])
```

One exception to the injunction above is when rendering a Quarto document to LaTeX using `theme_tt("multipage")"`. In that case, one must *avoid* using the Quarto chunk option, because these options trigger Quarto post-processing that will conflict with the `longtblr` environment used to split long tables across multiple pages.

The alternative is to use to refer to tables using standard LaTeX syntax: `\ref{tbl-ex-multipage}`. Then, use the `caption` argument in `tt()` to specify both the label and the caption:

```
tt(iris, caption = "Example table.\\label{tbl-ex-multipage}") |>
  theme_tt("multipage")
```

For standalone tables in any format (i.e., outside Quarto), you can use the `caption` argument like so:

```
tt(x, caption = "Blah blah.\\label{tbl-blah}")
```

1.4 Math

To insert LaTeX-style mathematical expressions in a `tinytable`, we enclose the expression in dollar signs: `$...$`. The expression will then rendered as a mathematical expression by MathJax (for HTML), LaTeX, or Pandoc. Do not forget to double escape any backslashes.

```
dat <- data.frame(Math = c(
  "$x^2 + y^2 = z^2$",
  "$\\frac{1}{2}$"
))
tt(dat) |> style_tt(j = 1, align = "c")
```

Math
$x^2 + y^2 = z^2$
$\frac{1}{2}$

In LaTeX (PDF), you can also use the `mode` inner setting from `tabularray` to render math in tables without delimiters (see Section 7.2 for details on `tabularray`):

```
dat <- data.frame(Math = c("x^2 + y^2 = z^2", "\\frac{1}{2}"))
tt(dat) |>
  style_tt(j = 1, align = "c", tabularray_inner = "column{1}={mode=math},")
```

Math
$x^2 + y^2 = z^2$
$\frac{1}{2}$

1.5 Line breaks and text wrapping

Manual line breaks work sligthly different in LaTeX (PDF) or HTML. This table shows the two strategies. For HTML, we insert a `<br>` tag. For LaTeX, we wrap the string in curly braces `{}`, and then insert two (escaped) backslashes: `\\\\`

```
d <- data.frame(
  "{Sed ut \\\\ perspiciatis unde}",
  "dicta sunt<br> explicabo. Nemo"
) |> setNames(c("LaTeX line break", "HTML line break"))
tt(d, width = 1)
```

LaTeX line break	HTML line break
Sed ut perspiciatis unde	dicta sunt explicabo. Nemo

1.6 Output formats

`tinytable` can produce tables in HTML, Word, Markdown, LaTeX, Typst, PDF, or PNG format. An appropriate output format for printing is automatically selected based on (1) whether the function is called interactively, (2) is called within RStudio, and (3) the output format of the Rmarkdown or Quarto document, if applicable. Alternatively, users can specify the print format in `print()` or by setting a global option:

```
tt(x) |> print("markdown")
tt(x) |> print("html")
tt(x) |> print("latex")

options(tinytable_print_output = "markdown")
```

With the `save_tt()` function, users can also save tables directly to PNG (images), PDF or Word documents, and to any of the basic formats. All we need to do is supply a valid file name with the appropriate extension (ex: `.png`, `.html`, `.pdf`, etc.):

```r
tt(x) |> save_tt("path/to/file.png")
tt(x) |> save_tt("path/to/file.pdf")
tt(x) |> save_tt("path/to/file.docx")
tt(x) |> save_tt("path/to/file.html")
tt(x) |> save_tt("path/to/file.tex")
tt(x) |> save_tt("path/to/file.md")
```

`save_tt()` can also return a string with the table in it, for further processing in `R`. In the first case, the table is printed to console with `cat()`. In the second case, it returns as a single string as an `R` object.

```r
tt(mtcars[1:10, 1:5]) |>
  group_tt(
    i = list(
      "Hello" = 3,
      "World" = 8),
    j = list(
      "Foo" = 2:3,
      "Bar" = 4:5)) |>
  print("markdown")
```

```
+------+-----+------+-----+------+
|      | Foo        | Bar        |
+------+-----+------+-----+------+
| mpg  | cyl | disp | hp  | drat |
+======+=====+======+=====+======+
| 21   | 6   | 160  | 110 | 3.9  |
+------+-----+------+-----+------+
| 21   | 6   | 160  | 110 | 3.9  |
+------+-----+------+-----+------+
| Hello                          |
+------+-----+------+-----+------+
| 22.8 | 4   | 108  | 93  | 3.85 |
+------+-----+------+-----+------+
| 21.4 | 6   | 258  | 110 | 3.08 |
+------+-----+------+-----+------+
| 18.7 | 8   | 360  | 175 | 3.15 |
+------+-----+------+-----+------+
| 18.1 | 6   | 225  | 105 | 2.76 |
+------+-----+------+-----+------+
| 14.3 | 8   | 360  | 245 | 3.21 |
```

```
+------+-----+------+-----+------+
| World                           |
+------+-----+------+-----+------+
| 24.4 | 4   | 147  | 62  | 3.69 |
+------+-----+------+-----+------+
| 22.8 | 4   | 141  | 95  | 3.92 |
+------+-----+------+-----+------+
| 19.2 | 6   | 168  | 123 | 3.92 |
+------+-----+------+-----+------+
```

```r
tt(mtcars[1:10, 1:5]) |>
  group_tt(
    i = list(
      "Hello" = 3,
      "World" = 8),
    j = list(
      "Foo" = 2:3,
      "Bar" = 4:5)) |>
  save_tt("markdown")
```

```
[1] "+------+-----+------+-----+------+\n|       | Foo         | Bar         |\n+------+-----+--
```

1.7 Combination and exploration

Tables can be explored, modified, and combined using many of the usual base `R` functions:

```r
a <- tt(mtcars[1:2, 1:2])
a
```

mpg	cyl
21	6
21	6

```r
dim(a)
```

```
[1] 2 2
```

```r
ncol(a)
```

```
[1] 2
```

```r
nrow(a)
```

```
[1] 2
```

```r
colnames(a)
```

```
[1] "mpg" "cyl"
```

Rename columns:

```r
colnames(a) <- c("a", "b")
a
```

a	b
21	6
21	6

Tables can be combined with the usual `rbind()` function:

```r
a <- tt(mtcars[1:3, 1:2], caption = "Combine two tiny tables.")
b <- tt(mtcars[4:5, 8:10])

rbind(a, b)
```

Table 3: Combine two tiny tables.

mpg	cyl	vs	am	gear
21	6			
21	6			
22.8	4			
		vs	am	gear
		1	0	3
		0	0	3

```r
rbind(a, b) |> format_tt(replace = "")
```

Table 4: Combine two tiny tables.

mpg	cyl	vs	am	gear
21	6			
21	6			
22.8	4			
		vs	am	gear
		1	0	3
		0	0	3

The `rbind2()` S4 method is slightly more flexible than `rbind()`, as it supports arguments `headers` and `use.names`.

Omit `y` header:

```r
rbind2(a, b, headers = FALSE)
```

Table 5: Combine two tiny tables.

mpg	cyl	vs	am	gear
21	6			
21	6			
22.8	4			
		1	0	3
		0	0	3

Bind tables by position rather than column names:

```r
rbind2(a, b, use_names = FALSE)
```

Table 6: Combine two tiny tables.

mpg	cyl	gear
21	6	
21	6	
22.8	4	
vs	am	gear
1	0	3
0	0	3

2 Formatting

```
library(tinytable)
options(tinytable_tt_digits = 3)
options(tinytable_theme_placement_latex_float = "H")
x <- mtcars[1:4, 1:5]
```

2.1 Numbers, dates, strings, etc.

The `tt()` function is minimalist; it's inteded purpose is simply to draw nice tables. Users who want to format numbers, dates, strings, and other variables in different ways should process their data *before* supplying it to the `tt()` table-drawing function. To do so, we can use the `format_tt()` function supplied by the `tinytable`.

In a very simple case—such as printing 2 significant digits of all numeric variables—we can use the `digits` argument of `tt()`:

```
dat <- data.frame(
     w = c(143002.2092, 201399.181, 100188.3883),
     x = c(1.43402, 201.399, 0.134588),
     y = as.Date(sample(1:1000, 3), origin = "1970-01-01"),
     z = c(TRUE, TRUE, FALSE))

tt(dat, digits = 2)
```

w	x	y	z
143002	1.43	1971-10-15	True
201399	201.4	1970-03-21	True
100188	0.13	1971-06-17	False

We can get more fine-grained control over formatting by calling `format_tt()` after `tt()`, optionally by specifying the columns to format with `j`:

```
tt(dat) |>
  format_tt(
    j = 2:4,
    digits = 1,
    date = "%B %d %Y") |>
  format_tt(
    j = 1,
    digits = 2,
    num_mark_big = " ",
    num_mark_dec = ",",
    num_fmt = "decimal")
```

w	x	y	z
143 002,21	1.4	October 15 1971	True
201 399,18	201.4	March 21 1970	True
100 188,39	0.1	June 17 1971	False

We can use a regular expression in `j` to select columns, and the `?sprintf` function to format strings, numbers, and to do string interpolation (similar to the `glue` package, but using Base `R`):

```
dat <- data.frame(
     a = c("Burger", "Halloumi", "Tofu", "Beans"),
     b = c(1.43202, 201.399, 0.146188, 0.0031),
     c = c(98938272783457, 7288839482, 29111727, 93945))
tt(dat) |>
  format_tt(j = "a", sprintf = "Food: %s") |>
  format_tt(j = 2, digits = 1) |>
  format_tt(j = "c", digits = 2, num_suffix = TRUE)
```

a	b	c
Food: Burger	1.432	99T
Food: Halloumi	201.399	7.3B
Food: Tofu	0.146	29M
Food: Beans	0.003	94K

Finally, if you like the `format_tt()` interface, you can use it directly with numbers, vectors, or data frames:

```
format_tt(pi, digits = 1)
```

```
[1] "3"
```

```
format_tt(dat, digits = 1, num_suffix = TRUE)
```

```
         a     b   c
1   Burger     1 99T
2 Halloumi   201  7B
3     Tofu   0.1 29M
4    Beans 0.003 94K
```

2.2 Significant digits and decimals

By default, `format_tt()` formats numbers to ensure that the smallest value in a vector (column) has at least a certain number of significant digits. For example,

```
k <- data.frame(x = c(0.000123456789, 12.4356789))
tt(k, digits = 2)
```

x
0.00012
12.43568

We can alter this behavior to ensure to round significant digits on a per-cell basis, using the `num_fmt` argument in `format_tt()`:

```r
tt(k) |> format_tt(digits = 2, num_fmt = "significant_cell")
```

x
0.00012
12

The numeric formatting options in `format_tt()` can also be controlled using global options:

```r
options("tinytable_tt_digits" = 2)
options("tinytable_format_num_fmt" = "significant_cell")
tt(k)
```

x
0.00012
12

2.3 Replacement

Missing values can be replaced by a custom string using the `replace` argument (default `""`):

```r
tab <- data.frame(a = c(NA, 1, 2), b = c(3, NA, 5))

tt(tab)
```

a	b
NA	3
1	NA
2	5

```r
tt(tab) |> format_tt()
```

a	b
	3
1	
2	5

```r
tt(tab) |> format_tt(replace = "-")
```

a	b
-	3
1	-
2	5

We can also specify multiple value replacements at once using a named list of vectors:

```r
tmp <- data.frame(x = 1:5, y = c(pi, NA, NaN, -Inf, Inf))
dict <- list("-" = c(NA, NaN), "-ϖ" = -Inf, "ϖ" = Inf)
tt(tmp) |> format_tt(replace = dict, digits = 2)
```

x	y
1	3.1
2	-
3	-
4	-∞
5	∞

2.4 Escape special characters

LaTeX and HTML use special characters to indicate strings which should be interpreted rather than displayed as text. For example, including underscores or dollar signs in LaTeX can cause compilation errors in some documents. To display those special characters, we need to substitute or escape them with backslashes, depending on the output format. The `escape` argument of `format_tt()` can be used to do this automatically:

```r
dat <- data.frame(
    "LaTeX" = c("Dollars $", "Percent %", "Underscore _"),
    "HTML" = c("<br>", "<sup>4</sup>", "<emph>blah</emph>")
)

tt(dat) |> format_tt(escape = TRUE)
```

LaTeX	HTML
Dollars $	
Percent %	⁴
Underscore _	<emph>blah</emph>

When applied to a `tt()` table, `format_tt()` will determine the type of escaping to do automatically. When applied to a string or vector, we must specify the type of escaping to apply:

```
format_tt("_ Dollars $", escape = "latex")
```

```
[1] "\\_ Dollars \\$"
```

2.5 Markdown

Markdown can be rendered in cells by using the `markdown` argument of the `format_tt()` function (note: this requires installing the `markdown` as an optional dependency).

```
dat <- data.frame( markdown = c(
  "This is _italic_ text.",
  "This sentence ends with a superscript.^2^")
)

tt(dat) |>
  format_tt(j = 1, markdown = TRUE) |>
  style_tt(j = 1, align = "c")
```

markdown
This is *italic* text.
This sentence ends with a superscript.^{2}

Markdown syntax can be particularly useful when formatting URLs in a table:

```
dat <- data.frame(
  `Package (link)` = c(
    "[`marginaleffects`](https://www.marginaleffects.com/)",
    "[`modelsummary`](https://www.modelsummary.com/)",
```

```r
    "[`tinytable`](https://vincentarelbundock.github.io/tinytable/)",
    "[`countrycode`](https://vincentarelbundock.github.io/countrycode/)",
    "[`WDI`](https://vincentarelbundock.github.io/WDI/)",
    "[`softbib`](https://vincentarelbundock.github.io/softbib/)",
    "[`tinysnapshot`](https://vincentarelbundock.github.io/tinysnapshot/)",
    "[`altdoc`](https://etiennebacher.github.io/altdoc/)",
    "[`plot2`](https://grantmcdermott.com/plot2/)",
    "[`parameters`](https://easystats.github.io/parameters/)",
    "[`insight`](https://easystats.github.io/insight/)"
  ),
  Purpose = c(
    "Interpreting statistical models",
    "Data and model summaries",
    "Draw beautiful tables easily",
    "Convert country codes and names",
    "Download data from the World Bank",
    "Software bibliographies in R",
    "Snapshots for unit tests using `tinytest`",
    "Create documentation website for R packages",
    "Extension of base R plot functions",
    "Extract from model objects",
    "Extract information from model objects"
  ),
  check.names = FALSE
)

tt(dat) |> format_tt(j = 1, markdown = TRUE)
```

Table 7: Vincent sometimes contributes to these R packages.

Package (link)	Purpose
marginaleffects	Interpreting statistical models
modelsummary	Data and model summaries
tinytable	Draw beautiful tables easily
countrycode	Convert country codes and names
WDI	Download data from the World Bank
softbib	Software bibliographies in R
tinysnapshot	Snapshots for unit tests using ‘tinytest‘
altdoc	Create documentation website for R packages
plot2	Extension of base R plot functions
parameters	Extract from model objects
insight	Extract information from model objects

2.6 Custom functions

On top of the built-in features of `format_tt`, a custom formatting function can be specified via the `fn` argument. The `fn` argument takes a function that accepts a single vector and returns a string (or something that coerces to a string like a number).

```
tt(x) |>
  format_tt(j = "mpg", fn = function(x) paste0(x, " mpg")) |>
  format_tt(j = "drat", fn = \(x) signif(x, 2))
```

mpg	cyl	disp	hp	drat
21 mpg	6	160	110	3.9
21 mpg	6	160	110	3.9
22.8 mpg	4	108	93	3.8
21.4 mpg	6	258	110	3.1

For example, the `scales` package which is used internally by `ggplot2` provides a bunch of useful tools for formatting (e.g. dates, numbers, percents, logs, currencies, etc.). The `label_*()` functions can be passed to the `fn` argument.

Note that we call `format_tt(escape = TRUE)` at the end of the pipeline because the column names and cells include characters that need to be escaped in LaTeX: `_`, `%`, and `$`. This last call is superfluous in HTML.

```
thumbdrives <- data.frame(
  date_lookup = as.Date(c("2024-01-15", "2024-01-18", "2024-01-14", "2024-01-16")),
  price = c(18.49, 19.99, 24.99, 24.99),
  price_rank = c(1, 2, 3, 3),
  memory = c(16e9, 12e9, 10e9, 8e9),
  speed_benchmark = c(0.6, 0.73, 0.82, 0.99)
)

tt(thumbdrives) |>
  format_tt(j = 1, fn = scales::label_date("%e %b", locale = "fr")) |>
  format_tt(j = 2, fn = scales::label_currency()) |>
  format_tt(j = 3, fn = scales::label_ordinal()) |>
  format_tt(j = 4, fn = scales::label_bytes()) |>
  format_tt(j = 5, fn = scales::label_percent())  |>
  format_tt(escape = TRUE)
```

date_lookup	price	price_rank	memory	speed_benchmark
2024-01-15	$18.49	1st	16 GB	60%
2024-01-18	$19.99	2nd	12 GB	73%
2024-01-14	$24.99	3rd	10 GB	82%
2024-01-16	$24.99	3rd	8 GB	99%

3 Style

The main styling function for the `tinytable` package is `style_tt()`. Via this function, you can access three main interfaces to customize tables:

1. A general interface to frequently used style choices which works for both HTML and LaTeX (PDF): colors, font style and size, row and column spans, etc. This is accessed through several distinct arguments in the `style_tt()` function, such as `italic`, `color`, etc.
2. A specialized interface which allows users to use the powerful `tabularray` package to customize LaTeX tables. This is accessed by passing `tabularray` settings as strings to the `tabularray_inner` and `tabularray_outer` arguments of `style_tt()`.

3. A specialized interface which allows users to use the powerful `Bootstrap` framework to customize HTML tables. This is accessed by passing CSS declarations and rules to the `bootstrap_css` and `bootstrap_css_rule` arguments of `style_tt()`.

These functions can be used to customize rows, columns, or individual cells. They control many features, including:

- Text color
- Background color
- Widths
- Heights
- Alignment
- Text Wrapping
- Column and Row Spacing
- Cell Merging
- Multi-row or column spans
- Border Styling
- Font Styling: size, underline, italic, bold, strikethrough, etc.
- Header Customization

The `style_*()` functions can modify individual cells, or entire columns and rows. The portion of the table that is styled is determined by the `i` (rows) and `j` (columns) arguments.

```
library(tinytable)
options(tinytable_tt_digits = 3)
options(tinytable_theme_placement_latex_float = "H")
x <- mtcars[1:4, 1:5]
```

3.1 Cells, rows, columns

To style individual cells, we use the `style_cell()` function. The first two arguments—`i` and `j`—identify the cells of interest, by row and column numbers respectively. To style a cell in the 2nd row and 3rd column, we can do:

```
tt(x) |>
  style_tt(
    i = 2,
    j = 3,
    background = "black",
    color = "white")
```

mpg	cyl	disp	hp	drat
21	6	160	110	3.9
21	6	160	110	3.9
22.8	4	108	93	3.85
21.4	6	258	110	3.08

The `i` and `j` accept vectors of integers to modify several cells at once:

```
tt(x) |>
  style_tt(
    i = 2:3,
    j = c(1, 3, 4),
    italic = TRUE,
    color = "orange")
```

mpg	cyl	disp	hp	drat
21	6	160	110	3.9
21	6	*160*	*110*	3.9
22.8	4	*108*	*93*	3.85
21.4	6	258	110	3.08

We can style all cells in a table by omitting both the `i` and `j` arguments:

```
tt(x) |> style_tt(color = "orange")
```

mpg	cyl	disp	hp	drat
21	6	160	110	3.9
21	6	160	110	3.9
22.8	4	108	93	3.85
21.4	6	258	110	3.08

We can style entire rows by omitting the `j` argument:

```r
tt(x) |> style_tt(i = 1:2, color = "orange")
```

mpg	cyl	disp	hp	drat
21	6	160	110	3.9
21	6	160	110	3.9
22.8	4	108	93	3.85
21.4	6	258	110	3.08

We can style entire columns by omitting the `i` argument:

```r
tt(x) |> style_tt(j = c(2, 4), bold = TRUE)
```

mpg	**cyl**	disp	**hp**	drat
21	**6**	160	**110**	3.9
21	**6**	160	**110**	3.9
22.8	**4**	108	**93**	3.85
21.4	**6**	258	**110**	3.08

The `j` argument accepts integer vectors, character vectors, but also a string with a Perl-style regular expression, which makes it easier to select columns by name:

```r
tt(x) |> style_tt(j = c("mpg", "drat"), color = "orange")
```

mpg	cyl	disp	hp	drat
21	6	160	110	3.9
21	6	160	110	3.9
22.8	4	108	93	3.85
21.4	6	258	110	3.08

```r
tt(x) |> style_tt(j = "mpg|drat", color = "orange")
```

mpg	cyl	disp	hp	drat
21	6	160	110	3.9
21	6	160	110	3.9
22.8	4	108	93	3.85
21.4	6	258	110	3.08

Here we use a "negative lookahead" to exclude certain columns:

```
tt(x) |> style_tt(j = "^(?!drat|mpg)", color = "orange")
```

mpg	cyl	disp	hp	drat
21	6	160	110	3.9
21	6	160	110	3.9
22.8	4	108	93	3.85
21.4	6	258	110	3.08

Of course, we can also call the `style_tt()` function several times to apply different styles to different parts of the table:

```
tt(x) |>
  style_tt(i = 1, j = 1:2, color = "orange") |>
  style_tt(i = 1, j = 3:4, color = "green")
```

mpg	cyl	disp	hp	drat
21	6	160	110	3.9
21	6	160	110	3.9
22.8	4	108	93	3.85
21.4	6	258	110	3.08

3.2 Colors

The `color` and `background` arguments in the `style_tt()` function are used for specifying the text color and the background color for cells of a table created by the `tt()` function. This argument plays a crucial role in enhancing the visual appeal and readability of the table,

whether it's rendered in LaTeX or HTML format. The way we specify colors differs slightly between the two formats:

For HTML Output:

- Hex Codes: You can specify colors using hexadecimal codes, which consist of a `#` followed by 6 characters (e.g., `#CC79A7`). This allows for a wide range of colors.
- Keywords: There's also the option to use color keywords for convenience. The supported keywords are basic color names like `black`, `red`, `blue`, etc.

For LaTeX Output:

- Hexadecimal Codes: Similar to HTML, you can use hexadecimal codes. However, in LaTeX, you need to include these codes as strings (e.g., `"#CC79A7"`).
- Keywords: LaTeX supports a different set of color keywords, which include standard colors like `black`, `red`, `blue`, as well as additional ones like `cyan`, `darkgray`, `lightgray`, etc.
- Color Blending: An advanced feature in LaTeX is color blending, which can be achieved using the `xcolor` package. You can blend colors by specifying ratios (e.g., `white!80!blue` or `green!20!red`).
- Luminance Levels: The `ninecolors` package in LaTeX offers colors with predefined luminance levels, allowing for more nuanced color choices (e.g., "azure4", "magenta8").

Note that the keywords used in LaTeX and HTML are slightly different.

```
tt(x) |> style_tt(i = 1:4, j = 1, color = "#FF5733")
```

mpg	cyl	disp	hp	drat
21	6	160	110	3.9
21	6	160	110	3.9
22.8	4	108	93	3.85
21.4	6	258	110	3.08

Note that when using Hex codes in a LaTeX table, we need extra declarations in the LaTeX preamble. See `?tt` for details.

3.3 Alignment

To align columns, we use a single character, or a string where each letter represents a column:

```r
dat <- data.frame(
  a = c("a", "aa", "aaa"),
  b = c("b", "bb", "bbb"),
  c = c("c", "cc", "ccc"))

tt(dat) |> style_tt(j = 1:3, align = "c")
```

a	b	c
a	b	c
aa	bb	cc
aaa	bbb	ccc

```r
tt(dat) |> style_tt(j = 1:3, align = "lcr")
```

a	b	c
a	b	c
aa	bb	cc
aaa	bbb	ccc

In LaTeX documents (only), we can use decimal-alignment:

```r
z <- data.frame(pi = c(pi * 100, pi * 1000, pi * 10000, pi * 100000))
tt(z) |>
    format_tt(j = 1, digits = 8, num_fmt = "significant_cell") |>
    style_tt(j = 1, align = "d")
```

pi
314.159 27
3141.5927
31 415.927
314 159.27

3.4 Font size

The font size is specified in em units.

```r
tt(x) |> style_tt(j = "mpg|hp|qsec", fontsize = 1.5)
```

mpg	cyl	disp	hp	drat
21	6	160	110	3.9
21	6	160	110	3.9
22.8	4	108	93	3.85
21.4	6	258	110	3.08

3.5 Spanning cells (merging cells)

Sometimes, it can be useful to make a cell stretch across multiple colums or rows, for example when we want to insert a label. To achieve this, we can use the `colspan` argument. Here, we make the 2nd cell of the 2nd row stretch across three columns and two rows:

```r
tt(x)|> style_tt(
  i = 2, j = 2,
  colspan = 3,
  rowspan = 2,
  align = "c",
  alignv = "m",
  color = "white",
  background = "black",
  bold = TRUE)
```

mpg	cyl	disp	hp	drat
21	6	160	110	3.9
21	**6**			3.9
22.8				3.85
21.4	6	258	110	3.08

Here is the original table for comparison:

```r
tt(x)
```

mpg	cyl	disp	hp	drat
21	6	160	110	3.9
21	6	160	110	3.9
22.8	4	108	93	3.85
21.4	6	258	110	3.08

Spanning cells can be particularly useful when we want to suppress redundant labels:

```
tab <- aggregate(mpg ~ cyl + am, FUN = mean, data = mtcars)
tab <- tab[order(tab$cyl, tab$am),]
tab
```

```
  cyl am      mpg
1   4  0 22.90000
4   4  1 28.07500
2   6  0 19.12500
5   6  1 20.56667
3   8  0 15.05000
6   8  1 15.40000
```

```
tt(tab, digits = 2) |>
  style_tt(i = c(1, 3, 5), j = 1, rowspan = 2, alignv = "t")
```

cyl	am	mpg
4	0	23
	1	28
6	0	19
	1	21
8	0	15
	1	15

3.6 Headers

The header can be omitted from the table by deleting the column names in the `x` data frame:

```r
k <- x
colnames(k) <- NULL
tt(k)
```

21.0	6	160	110	3.90
21.0	6	160	110	3.90
22.8	4	108	93	3.85
21.4	6	258	110	3.08

The first is row 0, and higher level headers (ex: column spanning labels) have negative indices like -1. They can be styled as expected:

```r
tt(x) |> style_tt(i = 0, color = "white", background = "black")
```

mpg	cyl	disp	hp	drat
21	6	160	110	3.9
21	6	160	110	3.9
22.8	4	108	93	3.85
21.4	6	258	110	3.08

When styling columns without specifying `i`, the headers are styled in accordance with the rest of the column:

```r
tt(x) |> style_tt(j = 2:3, color = "white", background = "black")
```

mpg	cyl	disp	hp	drat
21	6	160	110	3.9
21	6	160	110	3.9
22.8	4	108	93	3.85
21.4	6	258	110	3.08

3.7 Conditional styling

We can use the standard `which` function from Base `R` to create indices and apply conditional stying on rows. And we can use a regular expression in `j` to apply conditional styling on columns:

```r
k <- mtcars[1:10, c("mpg", "am", "vs")]

tt(k) |>
  style_tt(
    i = which(k$am == k$vs),
    background = "teal",
    color = "white")
```

mpg	am	vs
21	1	0
21	1	0
22.8	1	1
21.4	0	1
18.7	0	0
18.1	0	1
14.3	0	0
24.4	0	1
22.8	0	1
19.2	0	1

3.8 Vectorized styling (heatmaps)

The `color`, `background`, and `fontsize` arguments are vectorized. This allows easy specification of different colors in a single call:

```r
tt(x) |>
  style_tt(
    i = 1:4,
    color = c("red", "blue", "green", "orange"))
```

mpg	cyl	disp	hp	drat
21	6	160	110	3.9
21	6	160	110	3.9
22.8	4	108	93	3.85
21.4	6	258	110	3.08

When using a single value for a vectorized argument, it gets applied to all values:

```
tt(x) |>
  style_tt(
    j = 2:3,
    color = c("orange", "green"),
    background = "black")
```

mpg	cyl	disp	hp	drat
21	6	160	110	3.9
21	6	160	110	3.9
22.8	4	108	93	3.85
21.4	6	258	110	3.08

We can also produce more complex heatmap-like tables to illustrate different font sizes in em units:

```
# font sizes
fs <- seq(.1, 2, length.out = 20)

# headless table
k <- data.frame(matrix(fs, ncol = 5))
colnames(k) <- NULL

# colors
bg <- hcl.colors(20, "Inferno")
fg <- ifelse(as.matrix(k) < 1.7, tail(bg, 1), head(bg, 1))

# table
tt(k, width = .7, theme = "void") |>
  style_tt(j = 1:5, align = "ccccc") |>
  style_tt(
    i = 1:4,
    j = 1:5,
    color = fg,
    background = bg,
    fontsize = fs)
```

	0.5	0.9	1.3	1.7
0.2	0.6	1.0	1.4	1.8
0.3	0.7	1.1	1.5	1.9
0.4	0.8	1.2	1.6	2.0

3.9 Lines (borders)

The `style_tt` function allows us to customize the borders that surround eacell of a table, as well horizontal and vertical rules. To control these lines, we use the `line`, `line_width`, and `line_color` arguments. Here's a brief overview of each of these arguments:

- `line`: This argument specifies where solid lines should be drawn. It is a string that can consist of the following characters:
  - `"t"`: Draw a line at the top of the cell, row, or column.
  - `"b"`: Draw a line at the bottom of the cell, row, or column.
  - `"l"`: Draw a line at the left side of the cell, row, or column.
  - `"r"`: Draw a line at the right side of the cell, row, or column.
  - You can combine these characters to draw lines on multiple sides, such as `"tbl"` to draw lines at the top, bottom, and left sides of a cell.
- `line_width`: This argument controls the width of the solid lines in em units (default: 0.1 em). You can adjust this value to make the lines thicker or thinner.
- `line_color`: Specifies the color of the solid lines. You can use color names, hexadecimal codes, or other color specifications to define the line color.

Here is an example where we draw lines around every border ("t", "b", "l", and "r") of specified cells.

```
tt(x, theme = "void") |>
    style_tt(
        i = 0:3,
        j = 1:3,
        line = "tblr",
        line_width = 0.4,
        line_color = "orange")
```

mpg	cyl	disp	hp	drat
21	6	160	110	3.9
21	6	160	110	3.9
22.8	4	108	93	3.85
21.4	6	258	110	3.08

And here is an example with horizontal rules:

```
tt(x, theme = "void") |>
    style_tt(i = 0, line = "t", line_color = "orange", line_width = 0.4) |>
    style_tt(i = 0, line = "b", line_color = "purple", line_width = 0.2) |>
    style_tt(i = 4, line = "b", line_color = "orange", line_width = 0.4)
```

mpg	cyl	disp	hp	drat
21	6	160	110	3.9
21	6	160	110	3.9
22.8	4	108	93	3.85
21.4	6	258	110	3.08

```
dat <- data.frame(1:2, 3:4, 5:6, 7:8)
colnames(dat) <- NULL

tt(dat, theme = "void") |>
  style_tt(
    line = "tblr", line_color = "white", line_width = 0.5,
    background = "blue", color = "white")
```

1	3	5	7
2	4	6	8

3.10 Cell padding

There is no argument in `style_tt()` to control the padding of cells. Thankfully, this is easy to control using CSS and `tabularray` options:

```r
tt(x) |> style_tt(
    i = 1:5,
    bootstrap_css = "padding-right: .2em; padding-top: 2em;",
    tabularray_inner = "rowsep={2em}, colsep = {.2em}"
)
```

mpg	cyl	disp	hp	drat
21	6	160	110	3.9
21	6	160	110	3.9
22.8	4	108	93	3.85
21.4	6	258	110	3.08

3.11 Markdown and Word

Styling for Markdown and Word tables is more limited than for the other formats. In particular:

- The only supported arguments are: `bold`, `italic`, and `strikeout`.
- Headers inserted by `group_tt()` cannot be styled using the `style_tt()` function.

These limitations are due to the fact that there is no markdown syntax for the other options (ex: colors and background), and that we create Word documents by converting a markdown table to .docx via the Pandoc software.

One workaround is to style the group headers directly in their definition by using markdown syntax:

```r
mtcars[1:4, 1:4] |>
  tt() |>
  group_tt(i = list("*Hello*" = 1, "__World__" = 3)) |>
  print("markdown")
```

```
+------+-----+------+-----+
| mpg  | cyl | disp | hp  |
+======+=====+======+=====+
| *Hello*                |
+------+-----+------+-----+
| 21   | 6   | 160  | 110 |
+------+-----+------+-----+
| 21   | 6   | 160  | 110 |
+------+-----+------+-----+
| __World__              |
+------+-----+------+-----+
| 22.8 | 4   | 108  | 93  |
+------+-----+------+-----+
| 21.4 | 6   | 258  | 110 |
+------+-----+------+-----+
```

4 Groups and labels

```r
library(tinytable)
options(tinytable_tt_digits = 3)
options(tinytable_theme_placement_latex_float = "H")
x <- mtcars[1:4, 1:5]
```

The `group_tt()` function can label groups of rows (`i`) or columns (`j`).

4.1 Rows

The `i` argument accepts a named list of integers. The numbers identify the positions where row group labels are to be inserted. The names includes the text that should be inserted:

```r
dat <- mtcars[1:9, 1:8]

tt(dat) |>
  group_tt(i = list(
    "I like (fake) hamburgers" = 3,
    "She prefers halloumi" = 4,
    "They love tofu" = 7))
```

mpg	cyl	disp	hp	drat	wt	qsec	vs
21	6	160	110	3.9	2.62	16.5	0
21	6	160	110	3.9	2.88	17	0
I like (fake) hamburgers							
22.8	4	108	93	3.85	2.32	18.6	1
She prefers halloumi							
21.4	6	258	110	3.08	3.21	19.4	1
18.7	8	360	175	3.15	3.44	17	0
18.1	6	225	105	2.76	3.46	20.2	1
They love tofu							
14.3	8	360	245	3.21	3.57	15.8	0
24.4	4	147	62	3.69	3.19	20	1
22.8	4	141	95	3.92	3.15	22.9	1

We can style group rows in the same way as regular rows:

```r
tt(dat) |>
  group_tt(
    i = list(
      "I like (fake) hamburgers" = 3,
      "She prefers halloumi" = 4,
      "They love tofu" = 7)) |>
  style_tt(
    i = c(3, 5, 9),
    align = "c",
    color = "white",
    background = "gray",
    bold = TRUE)
```

mpg	cyl	disp	hp	drat	wt	qsec	vs
21	6	160	110	3.9	2.62	16.5	0
21	6	160	110	3.9	2.88	17	0
I like (fake) hamburgers							
22.8	4	108	93	3.85	2.32	18.6	1
She prefers halloumi							
21.4	6	258	110	3.08	3.21	19.4	1
18.7	8	360	175	3.15	3.44	17	0
18.1	6	225	105	2.76	3.46	20.2	1
They love tofu							
14.3	8	360	245	3.21	3.57	15.8	0
24.4	4	147	62	3.69	3.19	20	1
22.8	4	141	95	3.92	3.15	22.9	1

4.2 Columns

The syntax for column groups is very similar, but we use the `j` argument instead. The named list specifies the labels to appear in column-spanning labels, and the values must be a vector of consecutive and non-overlapping integers that indicate which columns are associated to which labels:

```
tt(dat) |>
  group_tt(
    j = list(
      "Hamburgers" = 1:3,
      "Halloumi" = 4:5,
      "Tofu" = 7))
```

Hamburgers			Halloumi			Tofu	
mpg	cyl	disp	hp	drat	wt	qsec	vs
21	6	160	110	3.9	2.62	16.5	0
21	6	160	110	3.9	2.88	17	0
22.8	4	108	93	3.85	2.32	18.6	1
21.4	6	258	110	3.08	3.21	19.4	1
18.7	8	360	175	3.15	3.44	17	0
18.1	6	225	105	2.76	3.46	20.2	1
14.3	8	360	245	3.21	3.57	15.8	0
24.4	4	147	62	3.69	3.19	20	1
22.8	4	141	95	3.92	3.15	22.9	1

Here is a table with both row and column headers, as well as some styling:

```
dat <- mtcars[1:9, 1:8]
tt(dat) |>
  group_tt(
    i = list("I like (fake) hamburgers" = 3,
             "She prefers halloumi" = 4,
             "They love tofu" = 7),
    j = list("Hamburgers" = 1:3,
             "Halloumi" = 4:5,
             "Tofu" = 7)) |>
  style_tt(
    i = c(3, 5, 9),
    align = "c",
    background = "teal",
    color = "white") |>
  style_tt(i = -1, color = "teal")
```

Hamburgers			Halloumi			Tofu	
mpg	cyl	disp	hp	drat	wt	qsec	vs
21	6	160	110	3.9	2.62	16.5	0
21	6	160	110	3.9	2.88	17	0
I like (fake) hamburgers							
22.8	4	108	93	3.85	2.32	18.6	1
She prefers halloumi							
21.4	6	258	110	3.08	3.21	19.4	1
18.7	8	360	175	3.15	3.44	17	0
18.1	6	225	105	2.76	3.46	20.2	1
They love tofu							
14.3	8	360	245	3.21	3.57	15.8	0
24.4	4	147	62	3.69	3.19	20	1
22.8	4	141	95	3.92	3.15	22.9	1

We can also stack several extra headers on top of one another:

```
tt(x) |>
  group_tt(j = list("Foo" = 2:3, "Bar" = 5)) |>
  group_tt(j = list("Hello" = 1:2, "World" = 4:5))
```

Hello			World	
	Foo			Bar
mpg	cyl	disp	hp	drat
21	6	160	110	3.9
21	6	160	110	3.9
22.8	4	108	93	3.85
21.4	6	258	110	3.08

5 Themes

`tinytable` offers a very flexible theming framwork, which includes a few basic visual looks, as well as other functions to apply collections of transformations to `tinytable` objects in a

repeatable way. These themes can be applied by supplying a string or function to the `theme` argument in `tt()`. Alternatively, users can call the `theme_tt()` function.

The main difference between `theme_tt()` and the other options in package, is that whereas `style_tt()` and `format_tt()` aim to be output agnostic, `theme_tt()` supplies transformations that can be output-specific, and which can have their own sets of distinct arguments. See below for a few examples.

```
library(tinytable)
options(tinytable_tt_digits = 3)
options(tinytable_theme_placement_latex_float = "H")
x <- mtcars[1:4, 1:5]
```

5.1 Visual themes

To begin, let's explore a few of the basic looks supplied by themes:

```
tt(x, theme = "striped")
```

mpg	cyl	disp	hp	drat
21	6	160	110	3.9
21	6	160	110	3.9
22.8	4	108	93	3.85
21.4	6	258	110	3.08

```
tt(x) |> theme_tt("striped")
```

mpg	cyl	disp	hp	drat
21	6	160	110	3.9
21	6	160	110	3.9
22.8	4	108	93	3.85
21.4	6	258	110	3.08

```
tt(x, theme = "grid")
```

mpg	cyl	disp	hp	drat
21	6	160	110	3.9
21	6	160	110	3.9
22.8	4	108	93	3.85
21.4	6	258	110	3.08

```
tt(x, theme = "bootstrap")
```

mpg	cyl	disp	hp	drat
21	6	160	110	3.9
21	6	160	110	3.9
22.8	4	108	93	3.85
21.4	6	258	110	3.08

```
tt(x, theme = "void")
```

mpg	cyl	disp	hp	drat
21	6	160	110	3.9
21	6	160	110	3.9
22.8	4	108	93	3.85
21.4	6	258	110	3.08

5.2 Custom themes

Users can also define their own themes to apply consistent visual tweaks to tables. For example, this defines a themeing function and sets a global option to apply it to all tables consistently:

```
theme_vincent <- function(x, ...) {
  out <- x |>
    style_tt(color = "teal") |>
    theme_tt("placement")
  out@caption <- "Always use the same caption."
  return(out)
}
```

```r
options(tinytable_tt_theme = theme_vincent)

tt(mtcars[1:2, 1:2])
```

Table 8: Always use the same caption.

mpg	cyl
21	6
21	6

```r
tt(mtcars[1:3, 1:3])
```

Table 9: Always use the same caption.

mpg	cyl	disp
21	6	160
21	6	160
22.8	4	108

```r
options(tinytable_tt_theme = NULL)
```

5.3 Tabular

The `tabular` theme is designed to provide a more "raw" table, without a floating table environment in LaTeX, and without CSS or Javascript in HTML.

```r
tt(x) |> theme_tt("tabular") |> print("latex")
```

```
\begin{tblr}[          %% tabularray outer open
```

```
]                       %% tabularray outer close
{                       %% tabularray inner open
colspec={Q[]Q[]Q[]Q[]Q[]},
}                       %% tabularray inner close
\toprule
mpg & cyl & disp & hp & drat \\ \midrule %% TinyTableHeader
21   & 6 & 160 & 110 & 3.9  \\
21   & 6 & 160 & 110 & 3.9  \\
22.8 & 4 & 108 & 93  & 3.85 \\
21.4 & 6 & 258 & 110 & 3.08 \\
\bottomrule
\end{tblr}
```

5.4 Resize

The `resize` theme allows you to adjust the size of the table in LaTeX outputs, making it fit within a specified width of the page. This is useful for large tables that need to be scaled down to fit the document layout. This table will be scaled to 90% of the available line width, ensuring it fits nicely within the document.

```r
tmp <- cbind(mtcars, mtcars)[1:10,]

tt(tmp) |> theme_tt("resize", width = .9)
```

mpg	cyl	disp	hp	drat	wt	qsec	vs	am	gear	carb	mpg	cyl	disp	hp	drat	wt	qsec	vs	am	gear	carb
21	6	160	110	3.9	2.62	16.5	0	1	4	4	21	6	160	110	3.9	2.62	16.5	0	1	4	4
21	6	160	110	3.9	2.88	17	0	1	4	4	21	6	160	110	3.9	2.88	17	0	1	4	4
22.8	4	108	93	3.85	2.32	18.6	1	1	4	1	22.8	4	108	93	3.85	2.32	18.6	1	1	4	1
21.4	6	258	110	3.08	3.21	19.4	1	0	3	1	21.4	6	258	110	3.08	3.21	19.4	1	0	3	1
18.7	8	360	175	3.15	3.44	17	0	0	3	2	18.7	8	360	175	3.15	3.44	17	0	0	3	2
18.1	6	225	105	2.76	3.46	20.2	1	0	3	1	18.1	6	225	105	2.76	3.46	20.2	1	0	3	1
14.3	8	360	245	3.21	3.57	15.8	0	0	3	4	14.3	8	360	245	3.21	3.57	15.8	0	0	3	4
24.4	4	147	62	3.69	3.19	20	1	0	4	2	24.4	4	147	62	3.69	3.19	20	1	0	4	2
22.8	4	141	95	3.92	3.15	22.9	1	0	4	2	22.8	4	141	95	3.92	3.15	22.9	1	0	4	2
19.2	6	168	123	3.92	3.44	18.3	1	0	4	4	19.2	6	168	123	3.92	3.44	18.3	1	0	4	4

5.5 Placement

The `placement` theme offers control over the positioning of the table in LaTeX documents, using floating parameters like `H` (from the `float` LaTeX package) to specify where the table should appear.

```r
tt(x) |>
  theme_tt("placement", latex_float = "H") |>
  print(output = "latex")
```

```latex
\begin{table}[H]
\centering
\begin{tblr}[         %% tabularray outer open
]                     %% tabularray outer close
{                     %% tabularray inner open
colspec={Q[]Q[]Q[]Q[]Q[]},
}                     %% tabularray inner close
\toprule
mpg & cyl & disp & hp & drat \\ \midrule %% TinyTableHeader
21   & 6 & 160 & 110 & 3.9  \\
21   & 6 & 160 & 110 & 3.9  \\
22.8 & 4 & 108 & 93  & 3.85 \\
21.4 & 6 & 258 & 110 & 3.08 \\
\bottomrule
\end{tblr}
\end{table}
```

5.6 Multipage

The `multipage` theme is designed for LaTeX documents to allow long tables to continue across multiple pages. This theme ensures that tables are not truncated and that all data is presented clearly.

```r
tmp <- rbind(mtcars, mtcars)[, 1:6]

cap <- "A long 80\\% width table with repeating headers."

tt(tmp, width = .8, caption = cap) |>
    theme_tt("multipage", rowhead = 1)
```

Table 10: A long 80% width table with repeating headers.

mpg	cyl	disp	hp	drat	wt
21	6	160	110	3.9	2.62

Continued on next page

Table 10: A long 80% width table with repeating headers. (Continued)

mpg	cyl	disp	hp	drat	wt
21	6	160	110	3.9	2.88
22.8	4	108	93	3.85	2.32
21.4	6	258	110	3.08	3.21
18.7	8	360	175	3.15	3.44
18.1	6	225	105	2.76	3.46
14.3	8	360	245	3.21	3.57
24.4	4	146.7	62	3.69	3.19
22.8	4	140.8	95	3.92	3.15
19.2	6	167.6	123	3.92	3.44
17.8	6	167.6	123	3.92	3.44
16.4	8	275.8	180	3.07	4.07
17.3	8	275.8	180	3.07	3.73
15.2	8	275.8	180	3.07	3.78
10.4	8	472	205	2.93	5.25
10.4	8	460	215	3	5.42
14.7	8	440	230	3.23	5.34
32.4	4	78.7	66	4.08	2.2
30.4	4	75.7	52	4.93	1.61
33.9	4	71.1	65	4.22	1.83
21.5	4	120.1	97	3.7	2.46
15.5	8	318	150	2.76	3.52
15.2	8	304	150	3.15	3.44
13.3	8	350	245	3.73	3.84
19.2	8	400	175	3.08	3.85
27.3	4	79	66	4.08	1.94
26	4	120.3	91	4.43	2.14
30.4	4	95.1	113	3.77	1.51
15.8	8	351	264	4.22	3.17
19.7	6	145	175	3.62	2.77
15	8	301	335	3.54	3.57

Continued on next page

Table 10: A long 80% width table with repeating headers. (Continued)

mpg	cyl	disp	hp	drat	wt
21.4	4	121	109	4.11	2.78
21	6	160	110	3.9	2.62
21	6	160	110	3.9	2.88
22.8	4	108	93	3.85	2.32
21.4	6	258	110	3.08	3.21
18.7	8	360	175	3.15	3.44
18.1	6	225	105	2.76	3.46
14.3	8	360	245	3.21	3.57
24.4	4	146.7	62	3.69	3.19
22.8	4	140.8	95	3.92	3.15
19.2	6	167.6	123	3.92	3.44
17.8	6	167.6	123	3.92	3.44
16.4	8	275.8	180	3.07	4.07
17.3	8	275.8	180	3.07	3.73
15.2	8	275.8	180	3.07	3.78
10.4	8	472	205	2.93	5.25
10.4	8	460	215	3	5.42
14.7	8	440	230	3.23	5.34
32.4	4	78.7	66	4.08	2.2
30.4	4	75.7	52	4.93	1.61
33.9	4	71.1	65	4.22	1.83
21.5	4	120.1	97	3.7	2.46
15.5	8	318	150	2.76	3.52
15.2	8	304	150	3.15	3.44
13.3	8	350	245	3.73	3.84
19.2	8	400	175	3.08	3.85
27.3	4	79	66	4.08	1.94
26	4	120.3	91	4.43	2.14
30.4	4	95.1	113	3.77	1.51
15.8	8	351	264	4.22	3.17

Continued on next page

Table 10: A long 80% width table with repeating headers. (Continued)

mpg	cyl	disp	hp	drat	wt
19.7	6	145	175	3.62	2.77
15	8	301	335	3.54	3.57
21.4	4	121	109	4.11	2.78

6 Plots and images

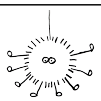
The `plot_tt()` function can embed images and plots in a `tinytable`. We can insert images by specifying their paths and positions (`i`/`j`).

```
library(tinytable)
options(tinytable_tt_digits = 3)
options(tinytable_theme_placement_latex_float = "H")
x <- mtcars[1:4, 1:5]
```

6.1 Inserting images in tables

To insert images in a table, we use the `plot_tt()` function. The `path_img` values must be relative to the main document saved by `save_tt()` or to the Quarto (or Rmarkdown) document in which the code is executed.

```
dat <- data.frame(
  Species = c("Spider", "Squirrel"),
  Image = ""
)

img <- c(
  "../figures/spider.png",
  "../figures/squirrel.png"
)

tt(dat) |>
  plot_tt(j = 2, images = img, height = 3)
```

Species	Image
Spider	
Squirrel	

In HTML tables, it is possible to insert tables directly from a web address, but not in LaTeX.

6.2 Inline plots

We can draw inline plots three ways, with

1. Built-in templates for histograms, density plots, and bar plots
2. Custom plots using base `R` plots.
3. Custom plots using `ggplot2`.

To draw custom plots, one simply has to define a custom function, whose structure we illustrate below.

6.2.1 Built-in plots

There are several types of inline plots available by default. For example,

```
plot_data <- list(mtcars$mpg, mtcars$hp, mtcars$qsec)

dat <- data.frame(
  Variables = c("mpg", "hp", "qsec"),
  Histogram = "",
  Density = "",
  Bar = "",
  Line = ""
)

# random data for sparklines
lines <- lapply(1:3, \(x) data.frame(x = 1:10, y = rnorm(10)))

tt(dat) |>
  plot_tt(j = 2, fun = "histogram", data = plot_data) |>
  plot_tt(j = 3, fun = "density", data = plot_data, color = "darkgreen") |>
```

```
  plot_tt(j = 4, fun = "bar", data = list(2, 3, 6), color = "orange") |>
  plot_tt(j = 5, fun = "line", data = lines, color = "blue") |>
  style_tt(j = 2:5, align = "c")
```

Variables	Histogram	Density	Bar	Line
mpg				
hp				
qsec				

6.2.2 Custom plots: Base R

Important: Custom functions must have `...` as an argument.

To create a custom inline plot using Base `R` plotting functions, we create a function that returns another function. `tinytable` will then call that second function internally to generate the plot.

This is easier than it sounds! For example:

```
f <- function(d, ...) {
  function() hist(d, axes = FALSE, ann = FALSE, col = "lightblue")
}

plot_data <- list(mtcars$mpg, mtcars$hp, mtcars$qsec)

dat <- data.frame(Variables = c("mpg", "hp", "qsec"), Histogram = "")

tt(dat) |>
  plot_tt(j = 2, fun = f, data = plot_data)
```

Variables	Histogram
mpg	
hp	
qsec	

6.2.3 Custom plots: `ggplot2`

Important: Custom functions must have `...` as an argument.

To create a custom inline plot using `ggplot2`, we create a function that returns a `ggplot` object:

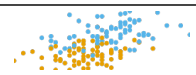
```
library(ggplot2)

f <- function(d, color = "black", ...) {
  d <- data.frame(x = d)
  ggplot(d, aes(x = x)) +
    geom_histogram(bins = 30, color = color, fill = color) +
    scale_x_continuous(expand=c(0,0)) +
    scale_y_continuous(expand=c(0,0)) +
    theme_void()
}

plot_data <- list(mtcars$mpg, mtcars$hp, mtcars$qsec)

tt(dat) |>
  plot_tt(j = 2, fun = f, data = plot_data, color = "pink")
```

Variables	Histogram
mpg	
hp	
qsec	

We can insert arbitrarily complex plots by customizing the `ggplot2` call:

```
penguins <- read.csv(
  "https://vincentarelbundock.github.io/Rdatasets/csv/palmerpenguins/penguins.csv",
  na.strings = "") |> na.omit()

# split data by species
dat <- split(penguins, penguins$species)
body <- lapply(dat, \(x) x$body_mass_g)
flip <- lapply(dat, \(x) x$flipper_length_mm)

# create nearly empty table
```

```r
tab <- data.frame(
  "Species" = names(dat),
  "Body Mass" = "",
  "Flipper Length" = "",
  "Body vs. Flipper" = "",
  check.names = FALSE
)

# custom ggplot2 function to create inline plot
f <- function(d, ...) {
  ggplot(d, aes(x = flipper_length_mm, y = body_mass_g, color = sex)) +
    geom_point(size = .2) +
    scale_x_continuous(expand=c(0,0)) +
    scale_y_continuous(expand=c(0,0)) +
    scale_color_manual(values = c("#E69F00", "#56B4E9")) +
    theme_void() +
    theme(legend.position = "none")
}

# `tinytable` calls
tt(tab) |>
  plot_tt(j = 2, fun = "histogram", data = body, height = 2) |>
  plot_tt(j = 3, fun = "density", data = flip, height = 2) |>
  plot_tt(j = 4, fun = f, data = dat, height = 2) |>
  style_tt(j = 2:4, align = "c")
```

Species	Body Mass	Flipper Length	Body vs. Flipper
Adelie			
Chinstrap			
Gentoo			

7 Customization

```r
library(tinytable)
options(tinytable_tt_digits = 3)
```

```r
options(tinytable_theme_placement_latex_float = "H")
x <- mtcars[1:4, 1:5]
```

7.1 HTML customization

The HTML customization options described in this section are not available for LaTeX (or PDF) documents. Please refer to the web documentation to read this part of the tutorial.

7.1.1 Bootstrap classes

7.1.2 CSS declarations

7.1.3 CSS rules

7.2 LaTeX / PDF customization

7.2.1 Preamble

Warning: Some of the features of this package may require a recent version of the `tabularray` package. Please update your local LaTeX distribution before using `tinytable`.

In Rmarkdown and Quarto documents, `tinytable` will automatically populate your LaTeX preamble with the necessary packages and commands. When creating your own LaTeX documents, you should insert these commands in the preamble:

```latex
\usepackage{tabularray}
\usepackage{float}
\usepackage{graphicx}
\usepackage[normalem]{ulem}
\UseTblrLibrary{booktabs}
\NewTableCommand{\tinytableDefineColor}[3]{\definecolor{#1}{#2}{#3}}
\newcommand{\tinytableTabularrayUnderline}[1]{\underline{#1}}
\newcommand{\tinytableTabularrayStrikeout}[1]{\sout{#1}}
```

7.2.2 Introduction to `tabularray`

`tabularray` offers a robust solution for creating and managing tables in LaTeX, standing out for its flexibility and ease of use. It excels in handling complex table layouts and offers enhanced functionality compared to traditional LaTeX table environments. This package is particularly useful for users requiring advanced table features, such as complex cell formatting, color management, and versatile table structures.

A key feature of Tabularray is its separation of style from content. This approach allows users to define the look and feel of their tables (such as color, borders, and text alignment) independently from the actual data within the table. This separation simplifies the process of formatting tables and enhances the clarity and maintainability of LaTeX code. The `tabularray` documentation is fantastic. It will teach you how to customize virtually every aspect of your tables: https://ctan.org/pkg/tabularray?lang=en

Tabularray introduces a streamlined interface for specifying table settings. It employs two types of settings blocks: Inner and Outer. The Outer block is used for settings that apply to the entire table, like overall alignment, while the Inner block handles settings for specific elements like columns, rows, and cells. The `style_tt()` function includes `tabularray_inner` and `tabularray_outer` arguments to set these respective features.

Consider this `tabularray` example, which illustrates the use of inner settings:

```
\begin{table}
\centering
\begin{tblr}[         %% tabularray outer open
]                     %% tabularray outer close
{                     %% tabularray inner open
column{1-4}={halign=c},
hlines = {bg=white},
vlines = {bg=white},
cell{1,6}{odd} = {bg=teal7},
cell{1,6}{even} = {bg=green7},
cell{2,4}{1,4} = {bg=red7},
cell{3,5}{1,4} = {bg=purple7},
cell{2}{2} = {r=4,c=2}{bg=azure7},
}                     %% tabularray inner close
mpg & cyl & disp & hp \\
21 & 6 & 160 & 110 \\
21 & 6 & 160 & 110 \\
22.8 & 4 & 108 & 93 \\
21.4 & 6 & 258 & 110 \\
18.7 & 8 & 360 & 175 \\
```

```
\end{tblr}
\end{table}
```

The Inner block, enclosed in `{}`, defines specific styles like column formats (`column{1-4}={halign=c}`), horizontal and vertical line colors (`hlines={fg=white}`, `vlines={fg=white}`), and cell colorations (`cell{1,6}{odd}={bg=teal7}`, etc.). The last line of the inner block also species that the second cell of row 2 (`cell{2}{2}`) should span 4 rows and 2 columns (`{r=4,c=3}`), be centered (`halign=c`), and with a background color with the 7th luminance level of the azure color (`bg=azure7`).

We can create this code easily by passing a string to the `tabularray_inner` argument of the `style_tt()` function:

```r
##| tbl-cap: "\\LaTeX{} table with colors and a spanning cell."
inner <- "
column{1-4}={halign=c},
hlines = {fg=white},
vlines = {fg=white},
cell{1,6}{odd} = {bg=teal7},
cell{1,6}{even} = {bg=green7},
cell{2,4}{1,4} = {bg=red7},
cell{3,5}{1,4} = {bg=purple7},
cell{2}{2} = {r=4,c=2}{bg=azure7},
"
mtcars[1:5, 1:4] |>
  tt(theme = "void") |>
  style_tt(tabularray_inner = inner)
```

mpg	cyl	disp	hp
21	6		110
21			110
22.8			93
21.4			110
18.7	8	360	175

7.2.3 `tabularray` keys

Inner specifications:

Key	Description and Values	Initial Value
`rulesep`	space between two hlines or vlines	`2pt`
`stretch`	stretch ratio for struts added to cell text	`1`
`abovesep`	set vertical space above every row	`2pt`
`belowsep`	set vertical space below every row	`2pt`
`rowsep`	set vertical space above and below every row	`2pt`
`leftsep`	set horizontal space to the left of every column	`6pt`
`rightsep`	set horizontal space to the right of every column	`6pt`
`colsep`	set horizontal space to both sides of every column	`6pt`
`hspan`	horizontal span algorithm: `default`, `even`, or `minimal`	`default`
`vspan`	vertical span algorithm: `default` or `even`	`default`
`baseline`	set the baseline of the table	`m`

Outer specifications:

Key	Description and Values	Initial Value
`baseline`	set the baseline of the table	`m`
`long`	change the table to a long table	None
`tall`	change the table to a tall table	None
`expand`	you need this key to use verb commands	None

Cells:

Key	Description and Values	Initial Value
`halign`	horizontal alignment: `l` (left), `c` (center), `r` (right) or `j` (justify)	`j`
`valign`	vertical alignment: `t` (top), `m` (middle), `b` (bottom), `h` (head) or `f` (foot)	`t`
`wd`	width dimension	None
`bg`	background color name	None
`fg`	foreground color name	None
`font`	font commands	None
`mode`	set cell mode: `math`, `imath`, `dmath` or `text`	None
`cmd`	execute command for the cell text	None
`preto`	prepend text to the cell	None
`appto`	append text to the cell	None
`r`	number of rows the cell spans	1
`c`	number of columns the cell spans	1

Rows:

Key	Description and Values	Initial Value
`halign`	horizontal alignment: `l` (left), `c` (center), `r` (right) or `j` (justify)	`j`
`valign`	vertical alignment: `t` (top), `m` (middle), `b` (bottom), `h` (head) or `f` (foot)	`t`
`ht`	height dimension	None
`bg`	background color name	None
`fg`	foreground color name	None
`font`	font commands	None
`mode`	set mode for row cells: `math`, `imath`, `dmath` or `text`	None
`cmd`	execute command for every cell text	None
`abovesep`	set vertical space above the row	`2pt`
`belowsep`	set vertical space below the row	`2pt`
`rowsep`	set vertical space above and below the row	`2pt`
`preto`	prepend text to every cell (like `>` specifier in `rowspec`)	None
`appto`	append text to every cell (like `<` specifier in `rowspec`)	None

Columns:

Key	Description and Values	Initial Value
`halign`	horizontal alignment: `l` (left), `c` (center), `r` (right) or `j` (justify)	`j`
`valign`	vertical alignment: `t` (top), `m` (middle), `b` (bottom), `h` (head) or `f` (foot)	`t`
`wd`	width dimension	None
`co`	coefficient for the extendable column (`X` column)	None
`bg`	background color name	None
`fg`	foreground color name	None
`font`	font commands	None
`mode`	set mode for column cells: `math`, `imath`, `dmath` or `text`	None
`cmd`	execute command for every cell text	None
`leftsep`	set horizontal space to the left of the column	`6pt`
`rightsep`	set horizontal space to the right of the column	`6pt`
`colsep`	set horizontal space to both sides of the column	`6pt`
`preto`	prepend text to every cell (like `>` specifier in `colspec`)	None
`appto`	append text to every cell (like `<` specifier in `colspec`)	None

hlines:

Key	Description and Values	Initial Value
`dash`	dash style: `solid`, `dashed` or `dotted`	`solid`
`text`	replace hline with text (like `!` specifier in `rowspec`)	None
`wd`	rule width dimension	`0.4pt`
`fg`	rule color name	None
`leftpos`	crossing or trimming position at the left side	`1`
`rightpos`	crossing or trimming position at the right side	`1`
`endpos`	adjust leftpos/rightpos for only the leftmost/rightmost column	`false`

vlines:

Key	Description and Values	Initial Value
`dash`	dash style: `solid`, `dashed` or `dotted`	`solid`
`text`	replace vline with text (like `!` specifier in `colspec`)	None
`wd`	rule width dimension	`0.4pt`
`fg`	rule color name	None
`abovepos`	crossing or trimming position at the above side	`0`
`belowpos`	crossing or trimming position at the below side	`0`

7.3 Shiny

`tinytable` is a great complement to Shiny for displaying HTML tables in a web app. The styling in a `tinytable` is applied by JavaScript functions and CSS. Thus, to ensure that this styling is preserved in a Shiny app, one strategy is to bake the entire page, save it in a temporary file, and load it using the `includeHTML` function from the `shiny` package. This approach is illustrated in this minimal example:

```
library("shiny")
library("tinytable")

fn <- paste(tempfile(), ".html")
tab <- tt(mtcars[1:5, 1:4]) |>
  style_tt(i = 0:5, color = "orange", background = "black") |>
  save_tt(fn)

shinyApp(
  ui = fluidPage(
    fluidRow(column(12, h1("This is test of tinytable"),
                    shiny::includeHTML(fn)))),
```

```
  server = function(input, output) {
  }
)
```

8 FAQ

8.1 LaTeX: `setspace`

Some users have encountered spacing issues when using `tinytable` to generate tables that are *not* wrapped in a `\begin{table}` environment (ex: `multipage` or raw `tblr`).

The `\begin{table}` resets any spacing commands in the preamble or body, such as:

```
\usepackage{setspace}
\doublespacing
```

This means that when using `theme_tt("multipage")` —which does not wrap the table in a `table` environment— tables are double spaced. One solution is to add the following to the document preamble when using multipage/longtblr:

```
\usepackage{etoolbox}
\AtBeginEnvironment{longtblr}{\begin{singlespacing}}
\AtEndEnvironment{longtblr}{\end{singlespacing}}
```

Example Quarto doc:

```
---
title: longtblr and setspacing
format:
  pdf:
    include-in-header:
      - text: |
         % Tinytable preamble
         \usepackage{tabularray}
         \usepackage{float}
         \usepackage{graphicx}
         \usepackage{codehigh}
         \usepackage[normalem]{ulem}
         \UseTblrLibrary{booktabs}
         \UseTblrLibrary{siunitx}
```

```
        \newcommand{\tinytableTabularrayUnderline}[1]{\underline
        {#1}}
        \newcommand{\tinytableTabularrayStrikeout}[1]{\sout{#1}}
        \NewTableCommand{\tinytableDefineColor}[3]{\definecolor{
        #1}{#2}{#3}}
        % Spacing Commands
        \usepackage{setspace}
        \doublespacing
        % Fix Spacing in longtblr
        \usepackage{etoolbox}
        \AtBeginEnvironment{longtblr}{\begin{singlespacing}}
        \AtEndEnvironment{longtblr}{\end{singlespacing}}
---

```{=latex}
\begin{longtblr}[         %% tabularray outer open
]                       %% tabularray outer close
{                       %% tabularray inner open
colspec={Q[]Q[]Q[]Q[]},
}                       %% tabularray inner close
\toprule
foo & bar & baz \\
foo & bar & baz \\
foo & bar & baz \\
\bottomrule
\end{longtblr}
```
```